Insurance application

Reference	**LP-APP-L047-W33**	Date	**2026-11-07 (data as of 2023-01-18, week 33)**
To	**Insurer (underwriting department)**	From	**Credit Risk, Banco Demo (fictional)**
Borrower	**Granja San Pedro S.A. (F045)**	Loan	**L047 · Crop input loan · $180,000**

Application for parametric index cover (drought, SPI-60) on field P047 (maize, 250 ha) for the 2023/24 season; sum insured $180,000, indicative premium $14,261 (7.92%). Submitted by the lender together with the applicant.

1. Applicant

Applicant (farmer)	**Granja San Pedro S.A.**	Farmer id	**F045**
Tax id (CUIT)	_____	Country / region	**Argentina / Buenos Aires · Norte**
Farm size	**327 ha (250 ha financed)**	Contact	_____

2. Loss payee and beneficiary

Banco Demo (fictional) (the lender) is named irrevocably as loss payee: any payout under this policy is assigned and paid directly to the lender to repay loan L047 (Crop input loan, $180,000, outstanding $180,000, maturity 17 Apr 2023); any balance is paid to the applicant.

3. Insured field

Field id	**P047**	Centroid (WGS84)	**lat -33.72380, lng -60.05000**
Boundary	**polygon of 5 vertices; first vertex lng -60.05221, lat -33.73450**	Area	**250 ha**
Crop	**maize**	Irrigated	**no**
Sowing date	**12 Oct 2022 (source: declared)**	Normal yield	**8.27 t/ha**

4. Product requested

Product	**Parametric index cover (drought, SPI-60)**
Index	**SPI-60: 60-day standardised precipitation index at the field (CHIRPS-type rainfall record, gamma fit over 20+ years)**
Trigger window	**Silking to Grain fill (11 to 19 weeks after sowing)**
Strike	**SPI-60 below -1.5: payout starts**
Exit	**SPI-60 at or below -2.5: full payout; linear between strike and exit**
Basis	**index only: no loss adjustment**

Trigger in words: SPI-60 < -1.5 between silking and grain fill.

5. Coverage period

Policy period	**11 Oct 2023 (expected sowing) to 28 Feb 2024 (expected harvest)**
Critical window	**27 Dec 2023 to 21 Feb 2024**

Existing cover **parametric (drought), sum insured $185,000: this application complements or replaces it**

6. Sum insured and premium

Item	Amount	Note
Sum insured	$180,000	= loan amount (default)
Expected annual payout (burn rate, 21 seasons)	$10,970	6.09% of the sum insured
Loading	x 1.3	expenses, risk margin, reinsurance
Indicative premium	**$14,261**	**7.92% of the sum insured**

Indicative only: the insurer prices with its own models; the premium is not binding until the policy is issued. The lender may finance the premium within the next loan.

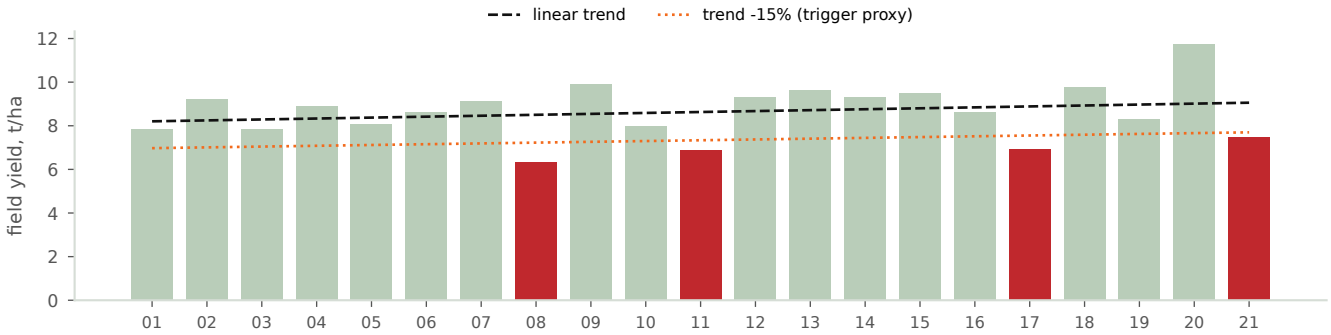
7. What this cover would have paid, season by season

Season	Field yield vs trend	Window min SPI-60	Payout
2022/23 (this season, to date)	-34% expected	**-3.09**	**$180,000**
2021/22	-18%	n/a	$22,720
2020/21	+30%	n/a	-
2019/20	-7%	n/a	-
2018/19	+9%	n/a	-
2017/18	-22%	n/a	$65,177
2016/17	-3%	n/a	-
2015/16	+8%	n/a	-
2014/15	+6%	n/a	-
2013/14	+10%	n/a	-
2012/13	+8%	n/a	-
2011/12	-21%	n/a	$49,628
2010/11	-7%	n/a	-
2009/10	+16%	n/a	-
2008/09	-25%	n/a	$92,841
2007/08	+8%	n/a	-
2006/07	+3%	n/a	-
2005/06	-4%	n/a	-
2004/05	+7%	n/a	-
2003/04	-5%	n/a	-
2002/03	+12%	n/a	-
2001/02	-4%	n/a	-

4 of 21 history seasons would have paid; burn rate 6.09% of the sum insured. Method note: the SPI record of past seasons is not stored in the analysis, so a past season counts as triggered when the field's yield fell more than 15% below its linear trend (full payout at -35%, linear between), a proxy for the index; 2022/23 uses the measured SPI-60 of this field.

8. Supporting evidence

Stress frequency: 4 of 21 history seasons (19%) had yield more than 15% below trend; worst 2008/09 at -25%. This season: Severe stress (score 100), expected yield -34% vs normal; lowest SPI-60 in the critical window -3.09. Median of the 5 monitored maize fields in Buenos Aires · Norte: window minimum SPI-60 -1.90.



9. Declarations

- The applicant confirms that the information on the field, crop, area and sowing date is true and complete, and authorises the lender and the insurer to verify it with satellite data.
- The applicant has not concealed any loss, claim or refusal of insurance on this field in the last five years.
- The applicant irrevocably assigns the proceeds of the policy to Banco Demo (fictional) up to the outstanding balance of loan L047.
- The applicant understands that index cover pays on the measured index, not on the actual loss on the farm (basis risk).
- The lender submits this application on the basis of data supplied by the applicant and of its own monitoring (synthetic demo data in this document).

10. Signatures

Applicant (farmer)	Lender (loss payee)	Insurer (received)
Name / date:	Name / date:	Name / date: